Market Reactions to Natural Disasters

Cyclone Ditwah's Impact on the Colombo Stock Exchange

Event Study Methodology — Complete Variable and Formula Guide

1. Your Raw Dataset — 66,958 Rows

Your dataset contains one row per stock per trading day. With 296 stocks over approximately 226 trading days, this produces roughly 66,958 rows. Every row feeds into the analysis — nothing is wasted.

Column	What It Represents	Example
symbol	Unique ticker code for the listed stock	DIAL
date	The calendar trading day	2025-03-01
high	Highest price the stock reached that day	15.00
low	Lowest price the stock reached that day	14.00
close	Final price when the market closed that day	14.50
volume	Total number of shares traded that day	50,000

2. The Two Time Windows

Before any formula is applied, your 226 trading days are split into two clearly defined zones. Everything is measured relative to Day 0 — 28 November 2025, the day Cyclone Ditwah made landfall.

2.1 Estimation Window — Day -120 to Day -6

This is the long normal period running from approximately February 2025 to late November 2025, well before any cyclone disruption began.

- **Purpose:** To teach the model what each stock's normal behavior looks like. These approximately 115 days per stock are used to learn how each stock typically responds to market movements.
- **Why it stops at Day -6:** Weather forecasts may have already started affecting prices in the final days before landfall. Stopping at Day -6 keeps this contamination out of the normal-behavior learning period.
- **Row count:** Approximately 115 days x 296 stocks = ~34,000 rows used here.

2.2 Event Window — Day -5 to Day +30

This is the period directly surrounding the cyclone — five trading days before landfall through thirty trading days after.

- **Why 5 days before:** To detect anticipation effects. Did the market start reacting to cyclone warnings before the storm hit?
- **Why 30 days after:** Recovery takes time. You need enough days to observe how long each sector took to return to normal.
- **Row count:** Approximately 36 days x 296 stocks = ~10,600 rows used here.

2.3 Visual Timeline

```
|-------- Estimation Window ----------|gap|--------- Event Window ----------|

Day -120                            Day -6  Day -5    Day 0             Day +30

Feb 2025                          ~20 Nov  ~21 Nov   28 Nov          ~16 Jan 2026

                                                        |

                                                Cyclone Landfall (Day 0)
```

Window	Day Range	Approx. Dates	Rows Used	Purpose
Estimation Window	Day -120 to Day -6	Feb 2025 - Nov 2025	~34,000	Learn normal stock behavior
Event Window	Day -5 to Day +30	21 Nov 2025 - 16 Jan 2026	~10,600	Detect abnormal behavior

Window	Day Range	Approx. Dates	Rows Used	Purpose
Remaining days	Outside both windows	—	Not used	Context only

3. Calculating Daily Returns — R(i,t) and R(m,t)

3.1 R(i,t) — Individual Stock Return

What it means: How much did stock i move on day t, expressed as a percentage change from the previous day?

```
R(i,t)  =  ( Close(today)  -  Close(yesterday) )  /  Close(yesterday)
```

Example for DIAL on 2025-03-02:

```
R(DIAL, 2025-03-02)  =  (14.20 - 14.50) / 14.50  =  -0.0207  =  -2.07%
```

Calculated for every row across all 226 days — not just the windows. You calculate all returns first, then split by window.

3.2 R(m,t) — Market Return (ASPI Index)

What it means: How much did the entire Colombo Stock Exchange move on day t? This is the benchmark representing the overall market.

```
R(m,t)  =  ( ASPI(today)  -  ASPI(yesterday) )  /  ASPI(yesterday)
```

One market return per day — the same value shared across all 296 stocks on that day. While R(i,t) is different for every stock, R(m,t) is identical for all stocks on a given date.

3.3 How Many Times R Values Are Calculated

Calculation	Times Done	Why
R(i,t) for DIAL	~225 times	One per day, every consecutive pair of days across 226 days. No value on Day 1 (no previous close).
R(m,t) for whole dataset	~225 times total	Once per day only — then reused for all 296 stocks on that day
R(i,t) for all 296 stocks	296 x 225 = 66,600 times	Every stock every day
R values used in estimation window	~115 per stock	Fed into Formula 1
R values used in event window	~36 per stock	Fed into Formula 2
R values outside both windows	~74 per stock	Not used in any formula

4. Formula 1 — The Market Model (Estimation Window)

Formula 1

R(i,t) = Alpha(i) + Beta(i) x R(m,t) + Epsilon(i,t)

Run once per stock using only the ~34,000 estimation window rows.

Method: Ordinary Least Squares — finds the best-fitting straight line through your data points

by minimising the total squared errors between actual and predicted values.

4.1 What Each Variable Means

Variable	Full Name	What It Means	How You Get It
R(i,t)	Stock return	How much stock i actually moved on day t	Calculated from close prices in Step 3
R(m,t)	Market return	How much the ASPI index moved on day t	Calculated from ASPI index in Step 3
Alpha(i)	Alpha — stock baseline	The stock's average return on days when the market return is exactly zero. Its own independent drift, unrelated to market movement.	Regression output — the intercept of the fitted line
Beta(i)	Beta — market sensitivity	How many percent the stock moves for every 1% the market moves. A fixed constant that does not change regardless of how big or small the market move is.	Regression output — the slope of the fitted line
Epsilon(i,t)	Epsilon — random noise	The random leftover the model cannot explain. Unpredictable day-to-day fluctuations.	Leftover error after fitting — not used further

4.2 Alpha and Beta in Plain English

Beta — The Market Follower

Beta answers: When the overall CSE market moves 1%, how much does this stock typically move?

Beta = 0.85 -> stock moves 0.85% when market moves 1% (calm, defensive)

Beta = 1.00 -> stock moves exactly with the market

Beta = 1.50 -> stock moves 1.50% when market moves 1% (aggressive, risky)

Beta = 0.20 -> stock barely reacts to market movements (very defensive)

Beta is learned from the estimation window. You are asking: historically,

how did this stock move relative to the market on normal days?

Alpha — The Stock's Own Drift

Alpha answers: On a day when the market does absolutely nothing — zero movement — what does this stock do on its own?

```
Alpha = +0.02%  ->  stock drifts up slightly on its own every day
Alpha =  0.00%  ->  stock has no independent movement
Alpha = -0.01%  ->  stock drifts down slightly on its own every day
```

Alpha captures everything about the stock not explained by market movement. In practice Alpha is usually very small — close to zero — because most of a stock's daily movement is explained by what the overall market does.

4.3 Why Linear Regression — And What Linearity Means

Linearity means Beta is a single fixed constant. The stock always moves that same proportion relative to the market regardless of how big or small the market move is. This creates a straight line relationship.

```
Market moves +1%  ->  Stock moves +0.85%

Market moves +2%  ->  Stock moves +1.70%

Market moves +3%  ->  Stock moves +2.55%

Market moves -1%  ->  Stock moves -0.85%
```

That constant proportional scaling is what linearity means. The formula has no R(m,t) squared, no logarithm, no interaction terms — just a straight multiplication of Beta by R(m,t).

Reason	Explanation
The goal is interpretability	You need to explain exactly what caused the abnormal return. Alpha and Beta give you that. A machine learning black box cannot.
The relationship actually is linear	Decades of financial research confirm the market model is approximately linear in normal conditions. You are not forcing a wrong model.
You want a clean residual	You are not trying to predict accurately — you want the unexplained gap. A deliberately simple stable baseline makes that gap meaningful.
It is the academic standard	MacKinlay (1997), cited in your proposal, established this as the gold standard. Reviewers expect it.
Sample size is appropriate	~115 estimation window days for 1 to 4 predictors is well within the required ratio. No overfitting risk.

4.4 Extending With Macro Variables

Your proposal also mentions exchange rate, interest rates, and inflation as control variables from the Central Bank of Sri Lanka and IMF data. These only affect Formula 1 — and only by adding more Beta terms.

```
Original:   R(i,t) = Alpha + Beta(market) x R(m,t) + Epsilon


Extended:   R(i,t) = Alpha + Beta(market) x R(m,t)

                           + Beta(fx)     x ExchangeRate(t)

                           + Beta(ir)     x InterestRate(t)

                           + Beta(inf)    x Inflation(t)

                           + Epsilon
```

Adding more Beta values makes the model more complex but the computer handles it identically — you just add more columns to the regression input. The downstream formulas (Formula 2 and 3) are completely unchanged. The only practical concern is multicollinearity — if exchange rate and inflation move together post-cyclone, the regression may struggle to separate their individual effects.

4.5 How Many Times Formula 1 Is Used

	For DIAL only	For all 296 stocks
Times Formula 1 runs	Once	296 times — once per stock
Input	115 estimation window rows for DIAL	~34,000 rows total
Output	1 Alpha value + 1 Beta value for DIAL	296 Alpha values + 296 Beta values

5. Formula 2 — Abnormal Return (Event Window)

Formula 2

AR(i,t) = R(i,t) - (Alpha-hat(i) + Beta-hat(i) x R(m,t))

Applied to each of the ~10,600 event window rows — one AR value per stock per event window day.

The hat symbol on Alpha and Beta means they were estimated from the estimation window.

What this formula does: For each stock on each event window day, it asks — given what the market did today, what should this stock have returned based on its normal behavior? The gap between what actually happened and what was expected is the abnormal return. This is the cyclone's fingerprint.

5.1 Which Window Does Formula 2 Use?

Formula 2 covers the entire event window — Day -5 to Day +30. This is both before AND after landfall, not just post-event.

Event Window Period	Days	What It Captures
Anticipation period	Day -5 to Day -1	Did the market react to cyclone warnings before landfall?
Landfall day	Day 0	The single-day shock when the cyclone hit
Immediate aftermath	Day +1 to Day +5	Panic selling and initial market response
Recovery period	Day +6 to Day +30	How stocks gradually returned toward normal

5.2 Worked Example — DIAL on Day 0

```
R(DIAL, Day 0)              = -3.50%   <-- DIAL actually fell 3.5% that day

Alpha-hat(DIAL)             = +0.02%   <-- baseline from estimation window

Beta-hat(DIAL)              =  0.85    <-- sensitivity from estimation window

R(m, Day 0)                 = -1.22%   <-- ASPI market fell 1.22% that day


Expected return  =  0.02  +  (0.85 x -1.22)  =  -1.02%

AR(DIAL, Day 0)  =  -3.50  -  (-1.02)        =  -2.48%
```

Interpretation

The -2.48% is purely the cyclone's effect on DIAL on landfall day.

The market's general decline already accounts for -1.02% of the drop.

The remaining -2.48% cannot be explained by normal market movement — it is the disaster's impact.

5.3 How Many Times Formula 2 Is Used

	For DIAL only	For all 296 stocks
Times Formula 2 runs	36 times — once per event window day	10,656 times (296 x 36)
Input per run	1 row: actual DIAL return + actual market return that day	1 row per stock per day
Output	1 AR value per day for DIAL	~10,600 AR values total
What changes each run	R(DIAL,t) and R(m,t) — one new day each time	Same — new day each time
What stays fixed	Alpha-hat and Beta-hat — never change	Same Alpha and Beta used every run

6. Formula 3 — Cumulative Abnormal Return (CAR)

Formula 3

CAR(i, t1, t2) = Sum of AR(i,t) for every day t between t1 and t2

Simply add up all daily abnormal returns for each stock across the chosen window.

What this formula does: Sums all the daily abnormal returns for a stock across a time period, giving one total number representing the cumulative damage or recovery caused by the cyclone.

6.1 Example for DIAL — Full Event Window

```
Day -5:  AR = -0.30%   <-- Market reacting to early cyclone warnings

Day -4:  AR = -0.10%

Day -3:  AR = +0.10%

Day -2:  AR = -0.50%

Day -1:  AR = -0.20%

Day  0:  AR = -2.48%   <-- Landfall day — biggest single-day drop

Day +1:  AR = -1.10%   <-- Continued selling pressure

Day +2:  AR = -0.80%

...

Day +15: AR = +0.30%   <-- Slow recovery begins

...

Day +30: AR = +0.20%



CAR(DIAL, -5, +30)  =  sum of all 36 daily AR values  =  -5.80%
```

DIAL lost 5.80% more than it should have over the entire event window — purely attributable to the cyclone.

6.2 Window Slicing — Different Questions from the Same Data

Window Slice	Days	Question It Answers
Pre-event	Day -5 to Day -1	Did stocks react before landfall? (anticipation effect)
Landfall day only	Day 0	What was the single-day shock on impact day?
First week	Day 0 to Day +5	How severe was the immediate aftermath?

Window Slice	Days	Question It Answers
Full window	Day -5 to Day +30	What was the total abnormal loss across the whole period?
Recovery phase	Day +10 to Day +30	When and how strongly did stocks begin recovering?

6.3 How Many Times Formula 3 Is Used

	For DIAL only	For all 296 stocks
Times Formula 3 runs	Once	296 times — once per stock
Input	36 AR values for DIAL	All ~10,600 AR values grouped by stock
Output	1 CAR value for DIAL — e.g. -5.80%	296 CAR values total

7. Complete Calculation Count — DIAL vs All 296 Stocks

Step	Calculation	Times for DIAL	Times for All 296 Stocks
Step 3	R(i,t) stock return calculated	~225 times	~66,600 times
Step 3	R(m,t) market return calculated	~225 times (shared)	~225 times total — reused by all stocks
Formula 1	Regression run	1 time (uses 115 rows)	296 times
Formula 1	Alpha values produced	1 Alpha for DIAL	296 Alpha values
Formula 1	Beta values produced	1 Beta for DIAL	296 Beta values
Formula 2	Abnormal return calculated	36 times (one per event day)	10,656 times
Formula 2	AR values produced	36 AR values for DIAL	~10,600 AR values total
Formula 3	CAR calculated	1 time	296 times
Formula 3	CAR values produced	1 CAR for DIAL	296 CAR values total

8. Sector Grouping

After computing 296 individual CAR values, you tag each stock with its CSE sector and group them. Sectors only enter the analysis at this point — not before.

```
sector_map = {

    'COMB' : 'Banking',

    'HAYL' : 'Agriculture',

    'EDEN' : 'Tourism',

    'DIAL' : 'Telecommunications',

    ... (all 296 stocks mapped to their CSE sector)

}
```

Sector	Individual Stock CARs	Sector Average CAR
Agriculture	-8.1%, -7.3%, -9.2%, -6.8%, ...	-7.90%
Tourism	-5.1%, -6.3%, -4.8%, ...	-5.40%
Banking	-1.2%, -0.8%, -1.5%, -2.1%, ...	-1.40%
Telecommunications	-0.5%, +0.2%, -0.3%, ...	-0.20%

You cannot just read these numbers and draw conclusions. You need statistical tests to

confirm whether the differences are genuine or random noise.

9. Statistical Testing — t-Test and Wilcoxon Signed-Rank Test

You have 296 CAR values — one per stock. You now need to answer two questions with statistical rigor:

- Was the overall market's abnormal return genuinely caused by the cyclone, or could it have occurred by random chance?
- Are the differences in CAR across sectors statistically real, or just noise?

9.1 The Cross-Sectional t-Test

What it is — in plain English

Imagine you have 48 agriculture stocks and each one has a CAR value.

The average CAR might be -7.9%. But is that meaningfully different from zero?

Maybe if you picked 48 random stocks on random days you would also get -7.9% by accident.

The t-test calculates a score that measures how far the average is from zero

relative to how spread out the individual stock CARs are.

A large t-score means the result is extremely unlikely to have happened by accident.

A small t-score means the average could easily be random noise.

How it is calculated:

```
t-score  =  Mean CAR  /  ( Standard Deviation of CARs  /  Square root of number of stocks
)
```

What the p-value means:

- p-value less than 0.05 -> Less than 5% chance this happened randomly. The cyclone caused this.
- p-value greater than 0.05 -> Could easily be random noise. Cannot confidently attribute to cyclone.

Worked example:

```
Agriculture sector — 48 stocks
Mean CAR       = -7.90%
Standard Dev   =  2.10%
t-score        = -7.90 / (2.10 / sqrt(48))  =  -26.1
p-value        =  0.0001
Conclusion: Agriculture stocks suffered a statistically significant abnormal loss.
```

Where it is used in your project:

- **Full market level:** Test whether the average CAR across all 296 stocks is significantly different from zero.
- **Per sector:** Test whether each sector's average CAR is significantly different from zero.
- **Comparing two sectors:** Test whether agriculture's CAR is significantly worse than banking's CAR.

9.2 The Wilcoxon Signed-Rank Test

What it is — in plain English

The t-test assumes your CAR values follow a roughly bell-shaped distribution.

In reality, stock returns around disasters are often skewed with extreme outliers.

The Wilcoxon test does not make this assumption. Instead of working with the actual CAR numbers, it ranks them from smallest to largest and tests whether the pattern of those ranks is consistent with a real effect — rather than random noise.

Think of it as a more robust backup test that confirms your t-test findings even when the data is not perfectly bell-shaped.

How it works step by step:

- Take all CAR values for a group — for example all 48 agriculture stocks.
- Rank them by absolute size from smallest to largest, ignoring whether positive or negative.
- Assign positive ranks to stocks with positive CARs and negative ranks to stocks with negative CARs.
- Sum the positive ranks and negative ranks separately.
- If the cyclone had no effect, both sums should be roughly equal. A large imbalance is evidence of a real effect.

9.3 Why You Use Both Tests Together

	Cross-Sectional t-Test	Wilcoxon Signed-Rank Test
Type	Parametric — assumes normal distribution	Non-parametric — no distribution assumption
Works on	The actual CAR values themselves	The ranks of the CAR values
Strength	Powerful when data is well-behaved	Robust when data has outliers or skew
Weakness	Can be misled by extreme outliers	Slightly less powerful when data is truly normal
Used for	Primary significance test	Confirmation and robustness check

Decision Rule for Reading Your Results

Both tests show $p < 0.05$ -> Strong, reliable conclusion. The effect is real.

Only t-test shows $p < 0.05$ -> Possibly driven by outliers. Report with caution.

Only Wilcoxon shows $p < 0.05$ -> Data may be skewed. Report both and explain.

Neither shows $p < 0.05$ -> This sector was not significantly affected by the cyclone.

9.4 Volatility and Liquidity Tests — Paired t-Test

The same paired t-test logic is applied to your High-Low spread and traded volume to measure how market conditions changed after the cyclone.

- **High-Low spread (volatility):** Compare average daily spread before the event window versus after landfall. A wider spread post-cyclone means the disaster increased market uncertainty and price swings.
- **Volume (liquidity):** Compare average daily volume before versus after. A significant drop means investors froze. A spike means panic selling.

```
Paired t-test:  Compare pre-event average  vs  post-event average
For each stock:  pre = average across estimation window,  post = average across event
window
```

10. What You Get in the End — Final Outputs

After passing all 66,958 rows through the three formulas, you end up with one core table and a set of answers to your research questions.

10.1 The Core Output Table — 296 Rows

Symbol	Sector	CAR(-5,+30)	CAR(-5,-1)	CAR(0,0)	CAR(0,+5)	CAR(0,+30)
DIAL	Telecom	-5.80%	-1.10%	-2.48%	-4.20%	-4.70%
COMB	Banking	-1.40%	-0.30%	-0.80%	-1.10%	-1.40%
HAYL	Agriculture	-9.20%	-1.80%	-3.50%	-7.10%	-9.20%
EDEN	Tourism	-6.30%	-1.20%	-2.90%	-5.00%	-6.10%
...	...	...	...	...	...	...

10.2 Questions Your Analysis Will Answer

Question	How Answered	Example Finding
Was the overall market genuinely affected?	Average all 296 CARs + t-test	Mean CAR = -4.2%, p = 0.0001. Yes, significant.

Question	How Answered	Example Finding
Which sectors were hit hardest?	Average CARs by sector + t-test per sector	Agriculture -7.9%, Tourism -5.4%, Banking -1.4%
Did the market anticipate the cyclone?	Look at CAR(-5,-1)	If significantly negative, market reacted to warnings
How bad was the single-day shock?	Look at CAR(0,0)	Mean CAR on Day 0 = -2.1% across all stocks
Did the market recover within 30 days?	Compare CAR(0,+5) vs CAR(0,+30)	If CAR shrinks toward zero, recovery happened
Did volatility increase?	Paired t-test on High-Low spread	Spread widened from 0.8% to 2.3%, p = 0.0003

10.3 The Most Important Visual Output

A chart showing each sector's CAR plotted day by day across the event window tells the entire story at a glance — which sectors the market punished, how quickly, and whether they recovered.

```
     0% |-------------------------------------------------
        |             Telecom .......................
 -2%    |       Banking --------
        |             Tourism ___________
 -4%    |
        |                  Construction ==============
 -6%    |
        |                       Agriculture ~~~~~~~~~~
 -8%    |
        |
        Day-5  Day-1  Day0  Day+5  Day+10  Day+20  Day+30
```

```
                    ^

                Landfall
```

11. The Complete Analysis Flow — All Variables Together

```
66,958 raw rows  (symbol, date, high, low, close, volume)

         |

         v

R(i,t): return per stock per day  ->  ~66,600 values

R(m,t): return of ASPI per day    ->  ~225 values  (shared across all stocks)

         |

         v

Split into time windows

         |

    _____|_____________

    |                 |

    v                 v

ESTIMATION WINDOW     EVENT WINDOW

~34,000 rows          ~10,600 rows
```

```
Day -120 to Day -6    Day -5 to Day +30

    |

    v

Formula 1  (Ordinary Least Squares regression — run once per stock)

  ->  296 Alpha values  +  296 Beta values

  ->  Optional: extend with exchange rate, interest rate, inflation Betas

              |

              v

Formula 2  (applied to every event window row)

  AR(i,t) = R(i,t) - ( Alpha-hat + Beta-hat x R(m,t) )

  ->  ~10,600 Abnormal Return values

              |

              v

Formula 3  (sum AR values per stock)

  CAR(i, t1, t2) = Sum of AR(i,t)

  ->  296 CAR values  (one total per stock)

              |

              v

Tag each stock with its CSE sector  ->  ~8 to 10 sector groups
```

```
              |
              v
Average CARs within each sector  ->  ~8 sector-level CAR averages
              |
              v
Cross-sectional t-test  +  Wilcoxon signed-rank test
  Full market:    Is overall CAR significantly different from zero?
  Per sector:     Is each sector's CAR significantly different from zero?
  Across sectors: Is agriculture significantly worse than banking?
              |
              v
Paired t-test on High-Low spread and Volume
  Did volatility and liquidity change significantly after the cyclone?
              |
              v
Conclusions:
  Which sectors were hit hardest and which were most resilient?
  Did the market anticipate the cyclone before landfall?
  How long did recovery take?
```

Does the CSE efficiently price natural disaster risk?

Group 6 | University of Moratuwa | 2025-2026